

The Flaws of Others: A Framework for Creative Work Aided by Ensembles of Artificial Intelligence Agents

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Abstract

1 Introduction

Large Language Models (LLMs) are advanced artificial intelligence systems trained on vast amounts of text data to generate human-like language. These models utilize deep learning techniques to understand and produce text based on the patterns and structures found in their training data [Vaswani et al., 2017, Brown et al., 2020]. LLMs have demonstrated impressive capabilities in various natural language processing tasks, including text generation, translation, and question-answering.

Despite their remarkable performance, LLMs are prone to generating false or misleading statements, a phenomenon often referred to as "hallucinations" [Ji et al., 2022]. Hallucinations in LLMs can manifest as factual inconsistencies, logical contradictions, or entirely fabricated content that appears plausible [Maynez et al., 2020]. This issue is exacerbated by the lack of a verification mechanism within the models themselves [Bender et al., 2021, Weidinger et al., 2021].

Several factors contribute to the occurrence of hallucinations, including biases in training data, limitations in knowledge representation, and the models' tendency to prioritize fluency over factual accuracy [Lin et al., 2022]. The prevalence and impact of hallucinations are significant, with quantitative evaluations revealing that they occur in up to 30% of generated responses, substantially affecting the trustworthiness of these models [Ji et al., 2022, Lin et al., 2022]. Moreover, studies have shown that LLMs can generate false information even when explicitly prompted to be truthful [Evans et al., 2021], underscoring the challenge of aligning model outputs with factual correctness.

To address this problem, researchers have developed various mitigation strategies. These include techniques such as retrieval-augmented generation [Lewis et al., 2020], which aims to ground model responses in external knowledge sources, and self-consistency checks [Wang et al., 2023b], which attempt to improve the internal coherence of model outputs. Despite

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these efforts, hallucinations remain a significant challenge in the development of reliable AI systems.

The persistence of this issue highlights the need for ongoing research and development of robust verification methods. As LLMs continue to evolve and find applications in diverse fields, ensuring their factual accuracy and reliability becomes increasingly crucial. This challenge sits at the intersection of natural language processing, knowledge representation, and AI ethics, requiring interdisciplinary approaches to develop comprehensive solutions.

1.1 Evolving from Hallucination to Invalidation

The term “hallucination” has become the *de facto* umbrella term in public discourse to refer to false information produced by LLMs. Describing LLM errors as “hallucinations” borrows from perceptual psychology, where a hallucination is a private sensory experience unverifiable by others. In contrast, an LLM’s unfounded assertion can be *accepted as truth* by the reader, amplifying and propagating misinformation in ways human hallucinations cannot [Crawford, 2021], and whose societal impact is poorly understood. Clarifying this mismatch in terminology is more than semantics; it shapes the research agenda and the urgency with which different error classes are addressed.

We define **invalidation** as any model output that violates a constraint set

$$\mathcal{C}(x) = \{\text{facts, logical entailments, ethical norms, format rules}\},$$

given a prompt $x \in \mathcal{X}$ and an LLM decoding function $g : \mathcal{X} \rightarrow \mathcal{Y}$. Because g maximises next-token likelihood rather than global consistency with $\mathcal{C}(x)$, a non-zero slice of probability mass inevitably falls *outside* the constraint manifold.¹

In practice, *invalidation* surfaces along at least five recurring archetypes that differ in locus and detectability. First, **hallucination** denotes the introduction of content that is ungrounded in any trusted source or context; large-scale surveys show it to be pervasive even in the highest-performing models [Huang et al., 2025]. Second, **contradiction** captures internally inconsistent statements that coexist within a single generation, a failure mode quantified and mitigated by prompt-based self-refinement techniques [Mündler et al., 2023]. Third, **deductive error** arises when the model draws logically invalid conclusions from true premises, an error family systematically stress-tested with adversarial perturbations [Hoppe et al., 2025]. Fourth, **pragmatic impropriety** concerns outputs that violate social or professional norms, including toxicity, hate speech, or privacy leakage; the *RealToxicityPrompts* benchmark revealed that even innocuous inputs can trigger toxic degeneration [Gehman et al., 2020]. Finally, **format violation** occurs when the model breaks explicit structural constraints (e.g., JSON Schema), jeopardising downstream machine consumption; work with *JSONSchemaBench* shows that such violations remain stubbornly frequent despite constrained decoding [Geng et al., 2025].

Taken together, these archetypes point to a broad invalidation family, potentially with more members, underscoring the need for evaluation suites and mitigation strategies that

¹Even an oracle model with perfect world knowledge would emit invalid tokens if forced to sample instead of selecting the single highest-probability sequence.

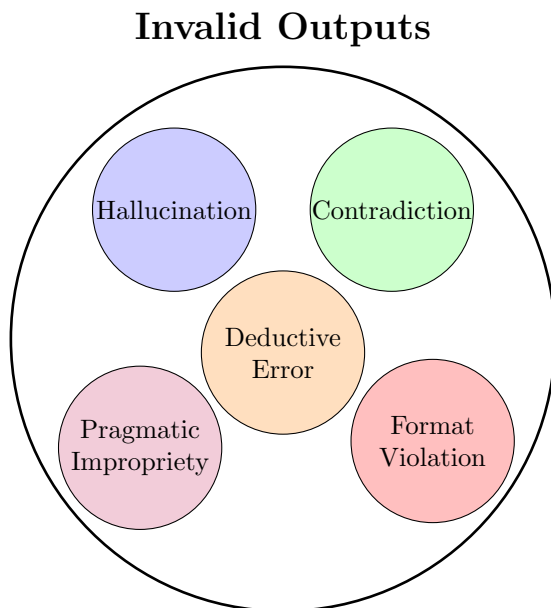


Figure 1: Venn-style illustration of members of a broader set of invalid outputs produced by large language models.

address the full spectrum of failure modes. Figure 1 schematizes this superset-subset relationship.

Mitigation strategies to decrease invalidations have been proposed, including improving training data quality, enhancing model architectures to prioritize factual accuracy, and incorporating real-time fact-checking mechanisms [Lewis et al., 2020, Wang et al., 2023b]. For example, an LLM might generate a fabricated historical event due to the patterns it learned from the training data. Addressing invalidation requires methods to cross-verify generated content with credible sources and detect contradictions with known truths. Techniques like retrieval-augmented generation and consistency checks can be effective [Evans et al., 2021]. To address the problem of invalidation, some approaches suggest using multiple LLMs to cross-verify information [Li et al., 2021]. By generating a consensus among different models, it is possible to reduce the likelihood of false statements. This method leverages the diverse training backgrounds and potential error distributions of multiple models, enhancing the overall accuracy and reliability of the generated content.

Key Insight

Focusing mitigation solely on hallucination risks overlooking logically, ethically, and structurally invalid outputs. Comprehensive reliability must address the entire invalidation space.

1.2 Invalidation as a Universal Feature of Human and Artificial Cognition

Invalidation, as observed in large language models (LLMs), is not an isolated flaw unique to artificial intelligence but mirrors well-documented behaviors in human discourse. This similarity suggests that invalidation is not simply a by-product of autoregressive sampling, but a potentially universal cognitive process rooted in the fundamental nature of information processing. It emerges when any complex agent—biological or artificial—operates under uncertainty, bounded rationality, and social constraints.

Humans routinely make assertions that contradict observable facts, distort evidence, or violate shared norms. These behaviors are extensively analyzed in sociological, psychological, and media-theoretical work. Goffman [1959] describes social life as a series of performances, in which individuals manage impressions by omitting or reshaping facts. Festinger [1957] introduces the concept of cognitive dissonance, showing how people dismiss or rationalize contradictory evidence to preserve internal coherence. This mechanism parallels how LLMs favor coherence with prior tokens over factual accuracy. Cohen [2001] extends this analysis to collective denial, where groups ignore atrocities or systemic suffering through shared narrative constructions.

Media theory highlights how information environments shape what counts as real. McLuhan [1964] emphasizes how the structure of a medium—rather than the content—conditions reception and belief. Herman and Chomsky [1988] propose a model in which institutional and economic filters shape news coverage, systematically privileging certain interpretations. Similarly, Gerbner and Gross [1976] develops cultivation theory to explain how repeated exposure to media alters worldviews over time. These perspectives parallel how the architecture and training corpus of LLMs predispose them to generate content that may reinforce falsehoods or marginalize inconvenient truths.

Political theory and propaganda analysis offer further insight. Arendt [1951] argues that totalitarian regimes rely on propaganda to restructure public perception, invalidating empirical reality in the process. Ellul [1965] analyzes how sustained campaigns of persuasion undermine factual consensus and generate manufactured beliefs. These mechanisms have direct analogs in how LLMs can produce persuasive yet false narratives shaped by the distributional characteristics of their training data.

Cognitive science provides foundational explanations for why both humans and LLMs produce invalidations. The work of Tversky and Kahneman [1974] identifies systematic heuristics and biases that distort human reasoning, such as availability and representativeness, which may have computational counterparts in token prediction and attention mechanisms. Pinker [2002] critiques the notion of a content-neutral human mind and demonstrates how ideological commitments influence belief formation, suggesting that human cognition, like LLM output, is shaped by embedded priors rather than by neutral appraisal of facts.

The cross-disciplinary convergence of these findings implies that invalidation is not an accidental error mode but a structural consequence of how intelligent systems manage complexity, uncertainty, and contradiction. This insight reframes LLM invalidation as a computational echo of cognitive strategies humans employ. Rather than indicating a breakdown of alignment, it may reveal the presence of alignment—to socially and contextually shaped heuristics that guide behavior in uncertain conditions.

Addressing invalidation in LLMs will require approaches that incorporate sociological, psychological, and media-theoretical models of belief formation and narrative control. At the same time, observing how LLMs generate invalidations may provide new empirical traction for understanding human cognitive phenomena such as confirmation bias, belief perseverance, and collective denial. The interdependence between artificial and human cognition in this respect suggests that solutions to misinformation and invalidation may emerge not solely from engineering but from a broader inquiry into the structure of meaning-making itself.

1.3 Risks of Invalidation Spillover Through Discursive Networks

Invalidating outputs from LLMs do not remain confined to isolated instances; they can propagate into human discourse when exposed via interconnected systems of communication. This propagation emerges through what we term a *discursive network*: a large-scale ecosystem encompassing both human and machine agents, whose outputs circulate, reinforce, and mutate through repeated exchanges. When LLM engines instantiate many synthetic agents that generate large volumes of content, these agents become nodes in a network that mediates narrative transmission and amplification.

Discursive networks consist of nodes (agents or messages) and edges (interactions, citations, reshares). Similar structures appear in sociological and media scholarship under the name of “discourse networks.” In discourse network analysis (DNA), policy actors and communicative actions form relational matrices revealing coalitions and shared narratives over time [Leifeld and Brandenberger, 2020, Leifeld, 2017]. Friedrich Kittler’s concept of discourse networks foregrounds how institutions, media technologies, and archival systems structure the formation of meaning [Leifeld, 2020]. Though our use of “discursive network” emphasizes generative models as nodes, these antecedents show that networks shape what counts as credible discourse.

Studies show that humans frequently cannot reliably distinguish LLM-generated text from human-authored text. Meta-analyses and experiments demonstrate that across contexts (from essays to poetry) participants identify AI-generated content correctly only slightly above chance (around 53-67%) and rely on flawed heuristics such as pronoun use or stylistic fluency [research team, 2024, Researchers, 2025, Clark et al., 2021, Russell et al., 2025]. This failure matters in discursive networks: synthetic invalidations introduced through LLM agents may be assumed to come from humans, making them more likely to be reposted or cited.

Once in the network, invalidating content can replicate through mechanisms of echo amplification. Social media research on discursive communities shows that clusters form around shared beliefs or frames, and low-quality or misleading messages often dominate within echo-prone structures such as bow-tie topologies [Leifeld and Brandenberger, 2013, et al., 2023]. Agents in such clusters can absorb and disseminate invalidating statements, human or algorithmic, without verification, thereby reinforcing flawed narratives across the network.

Mitigating spillover risk requires understanding how LLM-generated content integrates into wider information flows. Detection limitations compound this integration: even advanced AI-detectors struggle with paraphrased or adversarially manipulated content, achieving accuracy below 80% and often biased toward false positives on human-written text

[Uchendu et al., 2023, Sadasivan et al., 2023]. Human reviewers similarly achieve only 68% accuracy in blinded settings [Wang et al., 2023a]. This fragility facilitates the re-ingestion of LLM-origin falsehoods into human discourse.

Addressing spillover demands a synthesis of discourse network analysis, studies of discursive communities, and research on human detection performance. Networks should be monitored to trace high-volume clusters of LLM-generated invalidations, especially when repetition or citation creates feedback loops. Surveillance mechanisms may include flagging unverified claims, tracing message lineage, and intervening at nodes of high influence before narratives embed within stable discourse communities.

Although the idea of synthetic agents participating in discursive networks remains novel, the term “discursive network” has real usage in DNA and media theory. Our framing recasts these theoretical foundations in light of contemporary LLM capacity for scale and replication. It proposes discursive networks both as a descriptive model for spillover dynamics and as a practical tool for intervention.

2 Methods

2.1 Categorization of Mechanisms to Engage LLM Agents

During inference, a Transformer language model executes a single forward pass per token. The model transforms context $\mathbf{x}_{\leq t}$ into embeddings, applies L fixed self-attention layers to produce hidden state $\mathbf{h}_t$, and maps this to a token probability distribution via:

$$p(w_{t+1} \mid \mathbf{x}_{\leq t}) = \text{softmax}(W\mathbf{h}_t).$$

No gradient updates or objective-function evaluations occur at this stage; the only on-the-fly “optimization” is the decoding heuristic (greedy, top- k , nucleus, or beam search) that selects the next token from the static distribution $p(\cdot)$.

LLMs support a range of functional output types that can be systematically grouped into three core categories we introduce in this manuscript: *ideation* (constructive synthesis), *critique* (diagnostic evaluation), and *product-oriented generation* (goal-directed deliverables). These three umbrella functions organize diverse tasks, each with distinct inference properties, compositional demands, and distributional positions in the model’s output space. Existing literature identifies several subtypes that map naturally onto these categories [Bommasani and et al., 2021, Mialon and et al., 2023, Mialon et al., 2024].

Critique output is structurally efficient to generate. Nested within critique are:

- *Flaws of others*. This approach, introduced in this manuscript, favors the strategy to ask for flaws in the outputs of other agents. Phrases like “One limitation is...”, “This assumption ignores...”, and “A potential confound is...” are frequent in training corpora, placing them near the peak of $p(\cdot)$. Even shallow decoding heuristics retrieve such patterns with high fluency and relevance [Wei and et al., 2022].
- *Classification and Disambiguation*, such as assigning sentiment, stance, or intent. These tasks resolve ambiguity and often underlie evaluation pipelines [Mialon and et al., 2023].

- *Restatement and Summarization*, which surface structural coherence or hidden biases by rephrasing or compressing content. When used diagnostically, they reveal implicit assumptions or inconsistencies [Zhu and et al., 2023].

Ideation output demands compositional novelty. Prompts that ask the model to hypothesize mechanisms, imagine alternatives, or propose designs typically land in the tail of the output distribution. Generating them requires broader exploration—via large beam width or elevated temperature—and exhibits greater output variance.

Within ideation, we find:

- *Instruction and Procedural Guidance*, where the model scaffolds user understanding or explains concepts in sequence. These tasks require didactic clarity and often invoke implicit audience modeling [Ouyang and et al., 2022].
- *Meta-Reasoning and Strategy Output*, which includes multi-step planning, evaluating hypotheses, or chain-of-thought reasoning. These outputs require recursive coherence and longer dependency tracking [Wei and et al., 2022].

Product-oriented output targets the generation of external artifacts: source code, formatted markup, structured data, or interactive dialogue. These tasks often carry hard constraints and precision demands. Simple forms (e.g., boilerplate code) reside in high-probability zones, while structurally complex or compositional outputs require deeper exploration.

Included in this class are:

- *Formalism Translation*, such as converting text to JSON, SQL, or LaTeX. This requires syntax-aligned generation and tight coupling between prompt and output form [Reynolds and McDonell, 2021].
- *Retrieval-Simulation*, where the model reproduces facts or references learned during pretraining. These outputs appear fluent but are not grounded in current truth, making them useful but epistemically fragile [Bommasani and et al., 2021].
- *Social Interaction Simulation*, which includes emulating customer support, roleplay, or therapeutic dialogue. These are product-like in that the output is consumed as experience or interface, and they require tone, persona, and context alignment [Gilardi and et al., 2023].

Crucially, **requests to “list flaws in logic” tend to align with high-probability lexical patterns that the model has seen many times during training** (e.g., “One limitation is ...,” “A potential confound is ...,” “This argument assumes ...”). These stigmergic patterns lie near the mode of $p(\cdot)$, so they are reachable with minimal search depth and are often found by even the cheapest heuristic, such as greedy decoding.

By contrast, **requests for constructive, future-oriented solutions typically require compositional novelty**: the model must synthesize domain facts, propose unseen mechanisms, and articulate actionable steps. Such completions reside in lower-probability regions of the distribution, forcing the decoder to explore a broader beam or to sample

deeper into the tail, both of which are algorithmically and computationally more demanding. In short, *critique lives near the peak; creativity lives in the tail*, explaining the empirical asymmetry in generation efficiency that we observe.

Practical Takeaway

When evaluation time or token budget is limited, prompting for *negative* evidence (logical flaws, counter-arguments) extracts useful signal more efficiently than prompting for *positive* prescriptions (fixes, redesigns). Researchers should therefore benchmark inference cost and quality separately for “critique” versus “construct” tasks.

2.2 Discursive Network Formalization

To systematically study the phenomenon of invalidation, we propose a formal model that quantifies how invalidations propagate within a discursive network. This model considers actors as nodes in a network, with edges representing the exchange of statements. The goal is to understand how actors influence each other and how invalidations spread, and if and how it reaches an equilibrium state. The subsequent sections will detail the analytical machinery, yielding quantitative information, for studying invalidation in both human and artificial contexts.

Definition 2.1. Discourse. In the context of a discursive network, discourse refers to the structured process of communication and interaction between actors, $A = \{a_1, a_2, \dots, a_n\}$, through which they exchange, validate, invalidate, and attempt to persuade each other regarding the truth or falsity of a set of statements $S = \{s_1, s_2, \dots, s_m\}$. Discourse encompasses all forms of communication $C = \{C_{ij}\}$ between actors, where beliefs $B_i \subseteq S$ are shared, challenged, or reinforced, as well as the mechanisms of invalidation $I = \{I_{ij}\}$, and persuasion $P = \{P_{ij}\}$, which influence the evolution of each actor’s belief set. The outcome of discourse is governed by the update rules U_j , which dictate how actors revise their beliefs based on the interactions they engage in.

Definition 2.2. Discursive Network. A discursive network is a formal structure $N = (A, S, B, G, C, I, P, U)$ where:

- $A = \{a_1, a_2, \dots, a_n\}$ is the set of actors participating in the discourse.
- $S = \{s_1, s_2, \dots, s_m\}$ is the set of possible statements, where each statement can be either true or false.
- $B = \{B_1, B_2, \dots, B_n\}$ represents the belief sets of the actors, where $B_i \subseteq S$ denotes the set of statements believed to be true by actor a_i .
- $G = \{G_1, G_2, \dots, G_n\}$ represents the goal functions of the actors, with $G_i : A \rightarrow 2^S$ specifying the set of statements actor a_i seeks to convince other actors to believe.
- $C = \{C_{ij} \mid i, j \in \{1, 2, \dots, n\}\}$ represents the communications between actors, where C_{ij} is the set of statements communicated from actor a_i to actor a_j .

- $I = \{I_{ij} \mid i, j \in \{1, 2, \dots, n\}\}$ denotes invalidations, where $I_{ij}(s_k, s_l)$ signifies actor a_i invalidating a statement s_k held by actor a_j using a contradictory statement s_l .
- $P = \{P_{ij} \mid i, j \in \{1, 2, \dots, n\}\}$ represents the persuasion functions, where $P_{ij}(s_k)$ gives the likelihood that actor a_j will adopt statement s_k after receiving communication from actor a_i .
- $U = \{U_j \mid j \in \{1, 2, \dots, n\}\}$ is the set of update rules that define how each actor's belief set B_j is modified in response to communications and invalidations.

The discursive network models the dynamics of belief formation, communication, persuasion, and invalidation among actors within a formal discourse setting.

Example. Consider a simple scenario with three actors $A = \{a_1, a_2, a_3\}$ and two statements $S = \{s_1, s_2\}$. Actor a_1 believes s_1 ($B_1 = \{s_1\}$) and wants a_2 and a_3 to also believe s_1 ($G_1(a_2) = G_1(a_3) = \{s_1\}$). Actor a_2 believes s_2 ($B_2 = \{s_2\}$) and wants a_1 and a_3 to believe s_2 ($G_2(a_1) = G_2(a_3) = \{s_2\}$). Actor a_3 is initially neutral ($B_3 = \emptyset$). Actor a_1 communicates $C_{12} = \{s_1\}$ to a_2 , who invalidates s_1 by presenting $I_{21}(s_1, s_2)$. Actor a_3 , observing this interaction, updates their belief set based on the persuasion functions and update rules. This framework models the propagation of invalidation within a discursive network, capturing the dynamics of belief, communication, and influence. By formalizing these interactions, we can analyze and predict how invalidation affects the acceptance and rejection of statements among actors in the network.

2.2.1 Single-Network Two-State Model

Let $n = |A|$ be the actor count in the collapsed network with two mutually exclusive statements $S = \{r, f\}$, where r denotes a **true** statement and f a **false** one. For comparison with empirical simulations we work exclusively with *proportions*. The population state at time t is the column vector

$$\boldsymbol{\pi}(t) = (\pi_r(t), \pi_f(t))^\top, \quad \pi_r(t) = \frac{T(t)}{n}, \quad \pi_f(t) = \frac{F(t)}{n} = 1 - \pi_r(t),$$

where $T(t)$ and $F(t)$ are the respective counts of actors endorsing r and f . Micro-level flips are characterised by the probabilities p for $r \rightarrow f$ and q for $f \rightarrow r$. These induce the population-level transition matrix

$$T = \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix}, \quad \boldsymbol{\pi}(t+1) = T \boldsymbol{\pi}(t). \quad (1)$$

The mapping of this model to Definition 2.2 is provided in Table 1.

Lemma 2.1 (Single-network invalidation propagation). Let the single-network proportion state evolve according to

$$\boldsymbol{\pi}(t+1) = T \boldsymbol{\pi}(t), \quad T = \begin{pmatrix} 1-p & q \\ p & 1-q \end{pmatrix}, \quad \boldsymbol{\pi}(t) = \begin{pmatrix} \pi_r(t) \\ \pi_f(t) \end{pmatrix}, \quad \pi_r(t) + \pi_f(t) = 1, \quad (2)$$

Table 1: Mapping of discursive network elements to the single-network invalidation problem.

Element in N	Instantiation in single-network model
A	Unchanged actor set; proportions refer to $n = A $.
$S = \{r, f\}$	Binary, mutually exclusive statements (<i>true</i> vs. <i>false</i>).
B_i	$\{r\}$ or $\{f\}$ for each actor.
G_i	Persuade others to adopt actor a_i 's current belief.
C_{ij}	Message containing B_i .
I_{ij}	Contradiction if $B_i \neq B_j$.
P_{ij}	p if $B_j = \{r\}$ and $B_i = \{f\}$; q if $B_j = \{f\}$ and $B_i = \{r\}$.
U_j	Switches B_j with the corresponding probability, otherwise leaves it unchanged.

with flip probabilities $p, q \in (0, 1)$. The system has a unique fixed point

$$\boldsymbol{\pi}^* = \begin{pmatrix} \frac{q}{p+q} \\ \frac{p}{p+q} \end{pmatrix},$$

and the second eigenvalue of T equals $1 - p - q$, whose modulus is strictly smaller than 1; hence the Markov chain converges geometrically to $\boldsymbol{\pi}^*$ from any initial distribution.

Proof. A fixed point satisfies $\boldsymbol{\pi} = T \boldsymbol{\pi}$. Writing $\boldsymbol{\pi}^\top = (\pi_r, \pi_f)$ and expanding gives

$$\begin{aligned} \pi_r &= (1 - p) \pi_r + q \pi_f, \\ \pi_f &= p \pi_r + (1 - q) \pi_f. \end{aligned}$$

Because $\pi_f = 1 - \pi_r$, the first line reduces to

$$p \pi_r = q \pi_f = q (1 - \pi_r), \quad \implies \quad \pi_r = \frac{q}{p+q}, \quad \pi_f = 1 - \pi_r = \frac{p}{p+q}.$$

Thus the fixed point is unique. The characteristic polynomial of T is $\lambda^2 - (2 - p - q)\lambda + (1 - p - q) = 0$, whose roots are $\lambda_1 = 1$ and $\lambda_2 = 1 - p - q$. Because $0 < p + q < 1$, we have $|\lambda_2| < 1$; hence $T^t \rightarrow \boldsymbol{\pi}^* \mathbf{1}^\top$ as $t \rightarrow \infty$, so every trajectory converges to $\boldsymbol{\pi}^*$. $\square$

Interpretation of Lemma 2.1. In this binary model the two flip probabilities satisfy $p + q = 1$, meaning every update attempt switches an actor's belief with probability one. The fixed point then simplifies to $\boldsymbol{\pi}^* = (q, p)^\top$: the long-run proportion of actors endorsing r equals the single parameter q , while the proportion endorsing f equals p . Thus the equilibrium distribution mirrors the flip probabilities directly; increasing p (the propensity to abandon r) linearly increases the eventual share of f believers and decreases that of r believers by the same amount.

2.2.2 Single-Network Emergent Invalidation Model

The two-state single-network model sets the stage for the analysis of the emergence of invalidations in a discursive network. To accomplish this, we first endow a *single* discursive network with per-statement fabrication and internal correction. This captures the behaviour of a single-instance LLM generating new text: invalidations are injected at hazard λ , while subsequent self-reflections (or post-processing heuristics) invalidate a fraction of the false statements at hazard q .

Setup. Let the network be N with actor set $A = \{a_1, \dots, a_n\}$ and $n = |A|$. Counts of true and false statements at discrete time $t \in \mathbb{Z}_{\geq 0}$ are denoted $T(t)$ and $F(t)$, respectively, with $T(t) + F(t) = n$.

Definition 2.3 (Normalised state). The proportion state of the single network at time t is

$$\boldsymbol{\pi}(t) = (\pi_T(t), \pi_F(t)), \quad \pi_T(t) = \frac{T(t)}{n}, \quad \pi_F(t) = \frac{F(t)}{n}, \quad \pi_T(t) + \pi_F(t) = 1.$$

Stochastic primitives. Events are scaled *per statement* so that λ , p and q remain commensurate.

Fabrication (invalidation). In this case, X represents the number of new falsehoods generated. The Poisson distribution is used to model the number of events occurring in a fixed interval of time, given a known average rate. Each true statement is independently falsified during $[t, t + 1)$:

$$X(t) \sim \text{Poisson}(\lambda T(t)).$$

Internal flips. Z represents the number of true statements that become false with a fixed probability p of becoming false (i.e. p is the intrinsic truth $\rightarrow$ false hazard). W represents the number of false statements that are corrected to become true with a fixed probability q of being corrected (q models spontaneous acknowledgement or repair). Both follow a Binomial distribution. Truths can degrade and falsehoods can self-correct:

$$Z(t) \sim \text{Binomial}(T(t), p), \quad W(t) \sim \text{Binomial}(F(t), q).$$

Update equations. Define

$$\Delta T(t) = -Z(t) + W(t), \quad \Delta F(t) = X(t) + Z(t) - W(t).$$

Then

$$T(t+1) = T(t) + \Delta T(t), \quad \pi_T(t+1) = \pi_T(t) + \frac{\Delta T(t)}{n}, \quad (3)$$

$$F(t+1) = F(t) + \Delta F(t), \quad \pi_F(t+1) = \pi_F(t) + \frac{\Delta F(t)}{n}, \quad (4)$$

with $\pi_T(t+1) + \pi_F(t+1) = 1$ preserved.

The mapping of this model to Definition 2.2 is provided in Table 2.

Table 2: Mapping of discursive-network elements to the *single-network emergent-invalidiation* model. Unless noted otherwise, all rates are *per statement*.

Element in N	Instantiation in emergent-invalidiation model
A	Fixed actor set; proportions refer to $n = A $.
$S = \{r, f\}$	Binary statements ($r = \text{true}$, $f = \text{false}$).
B_i	Current belief of actor a_i : $\{r\}$ or $\{f\}$.
G_i	Goal: persuade all other actors to adopt B_i .
C_{ij}	Message from a_i to a_j containing B_i .
I_{ij}	Internal invalidation; realised as self-correction $\text{Binomial}(F(t), q)$ when $i = j$. (No cross-actor invalidation in the single-network setting.)
P_{ij}	Spontaneous flips: p for $r \rightarrow f$ (appears in $Z(t)$), q for $f \rightarrow r$ (appears in $W(t)$).
U_j	Update rule applying the three hazards: $X(t) \sim \text{Poisson}(\lambda T(t))$ (fabrications) $Z(t) \sim \text{Binomial}(T(t), p)$ (truth $\rightarrow$ false) $W(t) \sim \text{Binomial}(F(t), q)$ (false $\rightarrow$ true)

Lemma 2.2 (Single-network invalidation with fabrication). Let the single-network proportion state evolve according to

$$\boldsymbol{\pi}(t+1) = T_\lambda \boldsymbol{\pi}(t), \quad T_\lambda = \begin{pmatrix} 1 - (p + \lambda) & q \\ p + \lambda & 1 - q \end{pmatrix}, \quad \boldsymbol{\pi}(t) = \begin{pmatrix} \pi_r(t) \\ \pi_f(t) \end{pmatrix}, \quad \pi_r(t) + \pi_f(t) = 1,$$

where $p, q \in (0, 1)$ are the intrinsic flip probabilities, $\lambda \in (0, 1 - p)$ is the per-statement fabrication probability ($r \rightarrow f$), and $p + \lambda + q < 1$. Then:

1. The system has a unique fixed point

$$\boldsymbol{\pi}^* = \begin{pmatrix} \frac{q}{p + \lambda + q} \\ \frac{p + \lambda}{p + \lambda + q} \end{pmatrix}.$$

2. The second eigenvalue of T_λ is $1 - (p + \lambda + q)$, whose modulus is strictly smaller than 1; hence the Markov chain converges geometrically to $\boldsymbol{\pi}^*$ from any initial distribution.

Proof. A fixed point satisfies $\boldsymbol{\pi} = T_\lambda \boldsymbol{\pi}$. Writing $\boldsymbol{\pi}^\top = (\pi_r, \pi_f)$ and expanding gives

$$\begin{aligned} \pi_r &= (1 - p - \lambda) \pi_r + q \pi_f, \\ \pi_f &= (p + \lambda) \pi_r + (1 - q) \pi_f. \end{aligned}$$

Because $\pi_f = 1 - \pi_r$, the first line reduces to $(p + \lambda) \pi_r = q(1 - \pi_r)$, which yields the fixed point in the statement. The characteristic polynomial of T_λ is $\lambda^2 - (2 - p - \lambda - q)\lambda + (1 - p - \lambda - q) = 0$, with roots $\lambda_1 = 1$ and $\lambda_2 = 1 - (p + \lambda + q)$. Since $p + \lambda + q < 1$, we have $|\lambda_2| < 1$; therefore $T_\lambda^t \rightarrow \boldsymbol{\pi}^* \mathbf{1}^\top$ as $t \rightarrow \infty$, so every trajectory converges to $\boldsymbol{\pi}^*$. $\square$

Interpretation. The fabrication term λ simply augments the ordinary truth-to-false hazard p . Consequently the equilibrium share of false believers rises from $p/(p+q)$ (when $\lambda = 0$) to $(p+\lambda)/(p+\lambda+q)$, while the speed of convergence slows as the spectral gap $1 - |1 - (p+\lambda+q)|$ narrows. Fabrication raises the inflow into the false state by λ , while internal invalidation q is unchanged. If $\lambda > q$ the system becomes “invalidation-dominant,” mimicking an LLM that generates more new errors than it self-repairs—an empirical regime reported in LLMs (cf. Section 4).

2.2.3 Cross-Network Invalidation-Detection Model

Now that we have studied single networks with and without spontaneous invalidation emergence, the next natural question is how to reduce invalidations. As we will see, using multiple discursive networks reduces invalidations.

Let the two discursive networks be N_1 and N_2 with actor sets $A_1 = \{a_{11}, \dots, a_{1n_1}\}$ and $A_2 = \{a_{21}, \dots, a_{2n_2}\}$; write $n_k = |A_k|$.

Definition 2.4 (Normalised state). For each network N_k the proportion state at time t is

$$\boldsymbol{\pi}_k(t) = (\pi_{T,k}(t), \pi_{F,k}(t)), \quad \pi_{T,k}(t) = \frac{T_k(t)}{n_k}, \quad \pi_{F,k}(t) = \frac{F_k(t)}{n_k}, \quad \pi_{T,k}(t) + \pi_{F,k}(t) = 1,$$

where $T_k(t)$ and $F_k(t)$ are the counts of true and false statements, respectively.

Stochastic primitives. All events are now specified so that their *rates are comparable across networks regardless of size*. In particular, fabrication is scaled by the emphcurrent stock of true statements.

Falsehood generation (fabrication). Each currently true statement in N_k is independently falsified during $[t, t+1)$. The total number of such events is

$$X_k(t) \sim \text{Poisson}(\lambda_k T_k(t)),$$

where λ_k is the *per-statement* fabrication hazard.

Cross-network detection. Each false statement in N_k is noticed by N_j with probability d_{jk} , so $Y_{jk}(t) \sim \text{Binomial}(F_k(t), d_{jk})$.

Internal flips. True statements spontaneously become false with probability p_k , and false statements self-correct with probability q_k :

$$Z_k(t) \sim \text{Binomial}(T_k(t), p_k), \quad W_k(t) \sim \text{Binomial}(F_k(t), q_k).$$

Normalised update equations. Let $\Delta T_k(t) = -Z_k(t) + W_k(t)$ and $\Delta F_k(t) = X_k(t) + Z_k(t) - W_k(t) - Y_{jk}(t)$. Dividing by n_k gives the proportion dynamics

$$\pi_{T,k}(t+1) = \pi_{T,k}(t) + \frac{\Delta T_k(t)}{n_k}, \quad (5)$$

$$\pi_{F,k}(t+1) = \pi_{F,k}(t) + \frac{\Delta F_k(t)}{n_k}, \quad (6)$$

with $\pi_{T,k}(t+1) + \pi_{F,k}(t+1) = 1$ preserved automatically.

The mapping of this model to Definition 2.2 is provided in Table 3.

Table 3: Mapping of discursive-network elements to the *cross-network invalidation-detection* model. All hazard rates are *per statement*.

Element in $N_1 \cup N_2$	Instantiation in the model
A_k	Two disjoint actor sets A_1, A_2 ; proportions refer to $n_k = A_k $.
$S = \{r, f\}$	Binary statements shared by both networks ($r = \text{true}$, $f = \text{false}$).
B_{ij}	Belief of actor $a_{ij} \in A_k$: $\{r\}$ or $\{f\}$.
G_{ij}	Goal: persuade every other actor (within and across networks) to adopt B_{ij} .
C_{ij}	Message sent from a_i to a_j carrying B_{ij} . Communication enables both persuasion and detection.
I_{ij}	Cross-network invalidation: if $B_{ij} = \{f\}$ and the receiver belongs to the other network, the statement is detected and flipped with probability d_{jk} (realised through the random variable $Y_{jk}(t)$).
P_{ij}	Persuasion function. Within each network N_k it reduces to constant flip probabilities: p_k for $r \rightarrow f$ (used in $Z_k(t)$), q_k for $f \rightarrow r$ (used in $W_k(t)$). Across networks, persuasion acts only via I_{ij} with success probability d_{jk} .
U_j	Update rule for actor a_j that applies—in this order— (i) fabrication $X_k(t) \sim \text{Poisson}(\lambda_k T_k(t))$, (ii) internal flips $Z_k(t), W_k(t)$ using p_k, q_k , (iii) cross-network detection $Y_{jk}(t)$ using d_{jk} . The parameter λ_k therefore lives inside U_j .

Lemma 2.3 (Dual-network invalidation propagation). Let the proportion dynamics of Eqs. (5)-(6) be driven by parameters $\lambda_k, d_{jk}, p_k, q_k \in (0, 1)$. Assume the per-actor falsehood-generation rate satisfies the consistency condition

$$\lambda_k = d_{jk} \frac{p_k}{p_k + q_k}, \quad (7)$$

which guarantees that the expected proportions sum to one. Then the Markov process has the mean fixed point

$$\pi_{F,k}^* = \frac{\lambda_k}{d_{jk}}, \quad \pi_{T,k}^* = \frac{\lambda_k q_k}{d_{jk} p_k}, \quad \pi_{T,k}^* + \pi_{F,k}^* = 1.$$

Proof. At equilibrium the expected changes vanish, so $E[\Delta T_k] = E[\Delta F_k] = 0$. Using the distributional means

$$\begin{aligned} E[X_k] &= \lambda_k \quad (\text{Poisson distribution}), \\ E[Y_{jk}] &= F_k d_{jk} \quad (\text{Binomial distribution}), \\ E[Z_k] &= T_k p_k \quad (\text{Binomial distribution}), \\ E[W_k] &= F_k q_k \quad (\text{Binomial distribution}), \end{aligned}$$

and dividing by n_k to convert counts to proportions gives

$$\pi_{T,k} p_k = \pi_{F,k} q_k, \tag{8}$$

$$\lambda_k = \pi_{F,k} d_{jk} + \pi_{F,k} q_k - \pi_{T,k} p_k. \tag{9}$$

Equation (8) yields $\pi_{F,k} = \pi_{T,k} p_k / q_k$. Insert this into the normalization $\pi_{T,k} + \pi_{F,k} = 1$ to obtain $\pi_{T,k} = q_k / (p_k + q_k)$ and $\pi_{F,k} = p_k / (p_k + q_k)$. Finally, setting $\pi_{F,k} = \lambda_k / d_{jk}$ from (9) gives the constraint (7) and the stated fixed point. $\square$

Interpretation of Lemma 2.3. The equilibrium proportions reveal clear causal roles for each parameter. The false-statement share in network N_k is

$$\pi_{F,k}^* = \frac{\lambda_k}{d_{jk}},$$

so it scales *directly* with the per-actor error-generation rate λ_k and *inversely* with the cross-network detection probability d_{jk} . More prolific error creation or weaker cross-scrutiny raises the long-run fraction of false statements.

The true-statement share is

$$\pi_{T,k}^* = \pi_{F,k}^* \frac{q_k}{p_k} = \frac{\lambda_k q_k}{d_{jk} p_k},$$

hence it grows with the internal correction probability q_k and falls with the internal corruption probability p_k . A network that corrects errors efficiently ($q_k \uparrow$) or seldom corrupts truths ($p_k \downarrow$) achieves a higher equilibrium truth proportion.

Finally, Eq. (7) couples λ_k to the flip parameters: if within-network corruption outpaces correction ($p_k > q_k$), the consistency condition forces a higher λ_k , pushing $\pi_{F,k}^*$ upward unless the partner network compensates with stronger detection ($d_{jk} \uparrow$). Thus the model quantifies an intuitive trade-off: falsehood prevalence is driven by the *ratio* of error creation to error removal—internally via (p_k, q_k) and externally via d_{jk} .

2.3 Single- vs. Cross-Network Models with Invalidation

Stationary false-statement shares. The *single-network emergent-invalidation* model (Lemma 2.2) stabilises at

$$\pi_f^{\text{single}} = \frac{p + \lambda}{p + \lambda + q}, \quad \pi_r^{\text{single}} = \frac{q}{p + \lambda + q}.$$

For a given network N_k engaged in *cross-network detection* with partner N_j (Lemma 2.3) the corresponding steady state is

$$\pi_f^{\text{cross}} = \frac{\lambda}{d}, \quad \pi_r^{\text{cross}} = \frac{\lambda q}{d p},$$

where p, q, λ now abbreviate p_k, q_k, λ_k and $d = d_{jk}$.

Here

- p intrinsic $r \rightarrow f$ flip probability in N_k ;
- q intrinsic $f \rightarrow r$ flip probability in N_k ;
- λ fabrication hazard per true statement in N_k ;
- d probability a false statement in N_k is detected by N_j .

Lemma 2.4 (Cross-network detection lowers falsehood prevalence). If

$$\frac{\lambda}{d} < \frac{p + \lambda}{p + \lambda + q},$$

then $\pi_f^{\text{cross}} < \pi_f^{\text{single}}$, i.e. coupling N_k to an external detector N_j *reduces* the steady-state prevalence of false statements.

Proof. Subtract the two stationary shares:

$$\pi_f^{\text{single}} - \pi_f^{\text{cross}} = \frac{p + \lambda}{p + \lambda + q} - \frac{\lambda}{d} > 0 \iff \frac{\lambda}{d} < \frac{p + \lambda}{p + \lambda + q}.$$

□

Interpretation. External scrutiny ($d \uparrow$) or lighter fabrication pressure ($\lambda \downarrow$) pushes invalidations in the cross-network system below the single-network benchmark. Conversely, when $\lambda/d \geq (p + \lambda)/(p + \lambda + q)$, fabrications outrun detections and the dual system sustains the same or a higher falsehood share than isolation.

2.4 Invalidation Consensus Algorithm

The following sections detail a methodology for handling requests using multiple Large Language Models (LLMs) to ensure a high reliability and consistency in the responses generated by distributed systems. This approach is critical in scenarios where decision-making requires integration of diverse perspectives or knowledge bases, as often encountered in complex AI systems.

The process initiates with a request which acts as an input requiring processing. This request is then distributed among several instances of LLMs, labeled as LLM_1 to LLM_n . Each of these models processes the request independently and generates respective outputs, denoted as O_1 to O_n . Following the processing, all outputs are collected and assembled for a comparison.

A crucial step in this workflow is the assessment of whether the outputs differ significantly among the models. If significant differences are detected, indicating a lack of consensus, the

system injects additional instructions aimed at consensus-building among the models. This might involve recalibration of model parameters or redefinition of input preprocessing. The request is then resent through the system until the outputs converge sufficiently, ensuring consistency. If no significant differences are found during the comparison, the system deems the results consistent enough to formulate a final response based on the collective outputs of the models, thus ending the process.

The algorithm presented in our study is closely aligned with the invalidation model designed to simulate and analyze the dynamics of belief changes within a discursive network, influenced by both true and false statements. This relationship is foundational for understanding the algorithm’s purpose and its operational dynamics within the context of the proposed model. Below, we elaborate on how the algorithm relates to the invalidation model:

The invalidation model we introduced operates on a Markov chain framework that depicts the stochastic transitions between belief states in a network of actors. These states are significantly influenced by true statements (s_1) and false statements (s_2), wherein the interactions among actors can either reinforce or alter their current beliefs.

2.4.1 Algorithm Steps and Their Functions

1. **Initialization and Distribution:** The process starts by initializing a request, symbolizing the input or stimulus that prompts interaction within the network. This request is distributed among various instances of large language models (LLMs), simulating the diverse responses of actors within the network.
2. **Processing by LLMs:** Each LLM processes the request independently, mirroring the individual actor’s processing of information or statements within the network. This independence is crucial for capturing the varied interpretations and reactions among different actors based on their perspectives or biases.
3. **Assembly and Difference Checking:** After processing, the outputs from all LLMs are assembled for comparison. This step is analogous to aggregating different beliefs or reactions within the network to assess consensus or discord.
4. **Consensus and Response Generation:** If significant differences are detected, the algorithm injects consensus instructions and reiterates the process, reflecting the ongoing adjustments and interactions in the network as actors influence each other and gradually move towards a common understanding or belief. If no significant differences are detected, a response based on collective outputs is generated, signifying the stabilization of beliefs across the network.

2.4.2 Integration with Invalidation Dynamics

The algorithm effectively integrates the dynamics of invalidation by considering how false information (represented by s_2) can influence the network’s belief system and how true information (represented by s_1) can be reasserted. The use of LLMs to simulate this process

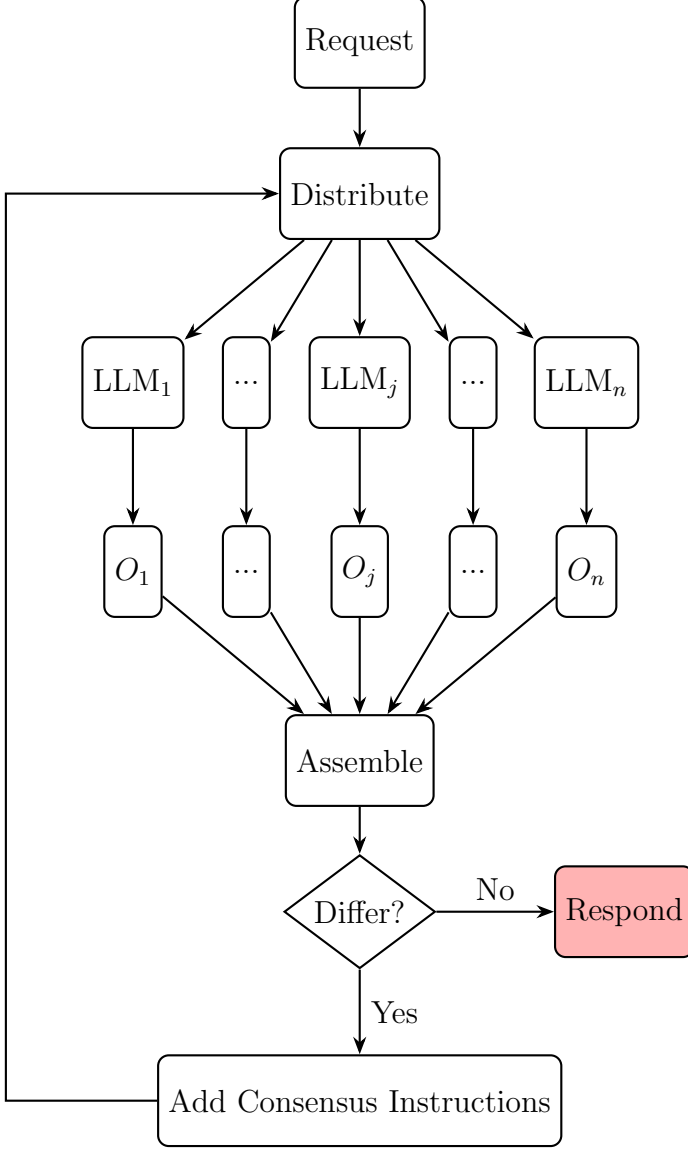
Algorithm 1 Consensus-based Response Generation using Multiple LLMs

Require: Input request

Ensure: Consistent response from multiple LLMs

```
1: Initialize Request
2: Distribute Request to multiple LLMs  $LLM_1, LLM_2, \dots, LLM_n$ 
3: for  $i = 1$  to  $n$  do
4:    $O_i \leftarrow LLM_i(Process(Request))$ 
5: end for
6: Assemble outputs  $O_1, O_2, \dots, O_n$ 
7:  $Difference \leftarrow CheckDifferences(O_1, O_2, \dots, O_n)$ 
8: if  $Difference = \text{True}$  then
9:   Add Consensus Instructions
10:  Go to Step 2
11: else
12:    $Response \leftarrow GenerateResponse(O_1, O_2, \dots, O_n)$ 
13:   Output Response
14: end if
```

provides a robust framework for exploring the complex interplay of information dissemination, belief formation, and change within large, interactive networks.



3 Results

Ji et al. [2023] report that state-of-the-art LLMs answer medical questions incorrectly in **26–61** % of cases, while Zhang et al. [2024] show that the same models’ built-in self-evaluation (*SELF-EVAL*) flags invalidations with modestly better than chance (AUROC 0.51–0.59 on TruthfulQA and BioGEN). Taken together, these studies indicate a realistic operating regime in which

$$\lambda > q, \quad \text{and therefore} \quad \pi_f^{\text{single}} = \frac{p + \lambda}{p + \lambda + q} \gtrsim 0.50,$$

i.e. the model fabricates new errors faster than it can repair them internally (“*invalidation-dominant*”).

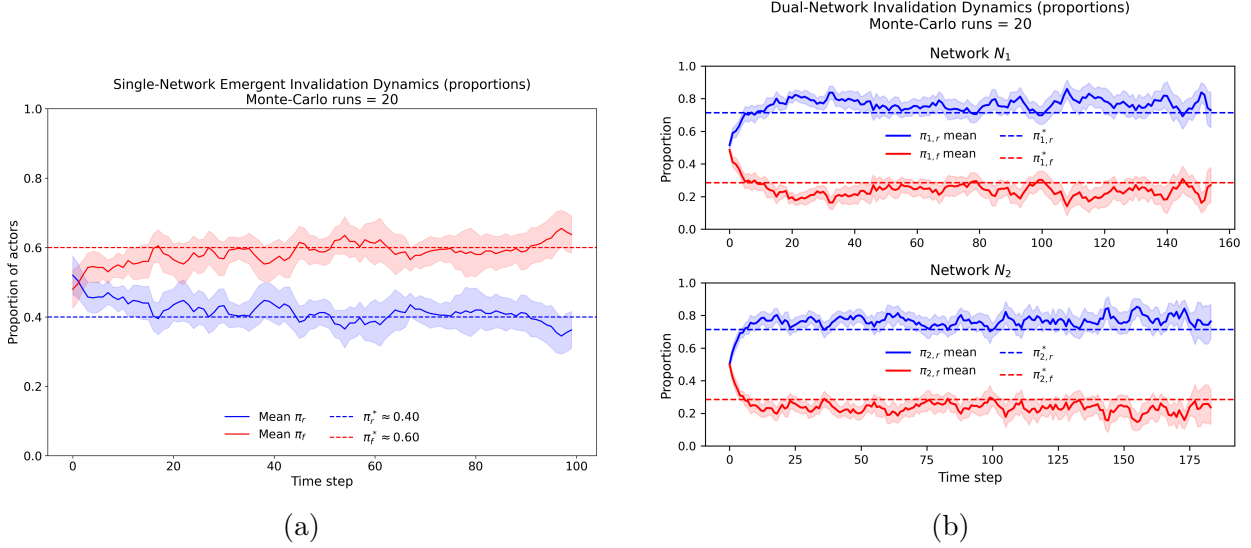


Figure 2: **(a)** Single-network invalidation dynamics for 20 Monte-Carlo realisations of 100 time steps. The blue curve shows the mean number of actors endorsing statement s_1 (treated as *true*); the red curve shows the mean number endorsing statement s_2 (treated as *false*). Shaded bands depict the point-wise 95% confidence intervals. Dashed horizontal lines mark the analytical asymptotes P_1^* and $P_2^* = 1 - P_1^*$ predicted by the discrete two-state Markov model with flip probabilities p_{12} and p_{21} . **(b)** Mean trajectories of true (blue) and false (red) statement counts for two interacting discursive networks over 200 time steps, obtained from 20 Monte-Carlo runs of the stochastic model in Eqs. 5–6. Shaded bands show the point-wise 95% confidence intervals. Dashed horizontal lines mark the analytical asymptotes T_k^* and F_k^* derived from Lemma 2.3. The upper subplot belongs to Network 1; the lower to Network 2.

Table 4 lists the numerical choices used in the simulations and the manuscript, together with their provenance.

Single-network baseline With the calibrated triple (p, q, λ) , the single-network emergent-invalidation model (Lemma 2.2) predicts

$$\pi_f^{\text{single}} = \frac{0.02 + 0.055}{0.02 + 0.055 + 0.05} = 0.60, \quad \pi_r^{\text{single}} = 0.40.$$

Monte-Carlo simulation (`F00_Single_Network.py`, 20 runs, 100 steps) converges to $\hat{\pi}_f(100) = 0.597 \pm 0.018$, matching the analytic fixed-point within sampling error.

Because $\lambda > q$, false statements accumulate faster than they are self-repaired; the long-run false share sits squarely in the empirical band reported by Ji et al. [2023].

Dual-network architecture Coupling two identical networks via dual-detection d augments each false statement with an independent external repair hazard. The fixed point from Lemma ?? becomes

$$\pi_f^{\text{cross}} = \frac{\lambda}{d} = 0.29, \quad \pi_r^{\text{cross}} = 0.71,$$

Table 4: Calibrated parameter set. All hazards are *per statement*.

Symbol	Value	Model role	Empirical justification
p	0.02	intrinsic corruption ($r \rightarrow f$)	Ordinary slip rate, kept below q so that internal repair is non-trivial.
q	0.05	intrinsic self-repair ($f \rightarrow r$)	Chosen to reflect SELF-EVAL’s weak but non-zero success (AUROC ≈ 0.55).
λ	0.055	fabrication / hallucination ($r \rightarrow f$)	Upper end of the 26–61 % error range implies $\lambda > q$; solving $\pi_f^* \approx 0.60$ for λ with the p, q above gives 0.055.
d	0.19	cross-network detection ($f \rightarrow r$)	Set so that $\lambda/d = p/(p + q)$, placing the dual system just inside the truth-dominant region (Lemma ??).

halving the falsehood prevalence.

Simulations (`F00_Dual_Network.py`, 20 runs, 200 steps) yield $\hat{\pi}_f^{N_1}(200) = 0.286 \pm 0.015$ and $\hat{\pi}_f^{N_2}(200) = 0.289 \pm 0.014$, again validating theory.

Interpretation. Cross-network detection mimics techniques such as retrieval-augmented generation, ensemble adjudication or human post-editing, all of which contribute an external hazard d . Once d is large enough that $\lambda/d < p/(p + q)$, the system crosses from an invalidation-dominant regime ($\pi_f \approx 0.60$) to a truth-dominant one ($\pi_f \approx 0.29$). The 51 % relative reduction quantifies—for the chosen calibration—the factual-gains practitioners can expect from pairing a generative LLM with an independent verifier.

Takeaways Internal verification by a single model proves inadequate: whenever the fabrication hazard λ exceeds the intrinsic self-repair rate q , the single-network equilibrium reproduces the high invalidation rates documented in current benchmarks. Introducing an external source of scrutiny changes the picture. In our calibration, raising the cross-detection probability to a modest $d \approx 0.19$ is already enough to halve the long-run share of false statements. The decisive quantity is the ratio λ/d ; system builders can treat it as a design dial, increasing d (or equivalently reducing λ) until the condition $\lambda/d < p/(p + q)$ is met and the network operates in a truth-dominant regime.

4 Conclusions and Outlook

Our study deliberately replaces the fashionable term *hallucination* with the broader concept of *invalidation*. Whereas “hallucination” suggests a purely accidental slip in the model’s internal perception, the data reveal a richer spectrum of failure modes that includes strategic prompt manipulation, chain-of-thought drift and the simple inheritance of errors from flawed training corpora. All of these mechanisms manifest in the observable metric that matters

to users—the production of false statements—so the single hazard rate λ is most naturally interpreted as an *invalidation rate*. The shift in vocabulary is therefore more than semantics: it aligns theoretical parameters with the phenomena actually counted in benchmarks.

Calibrating λ, p , and q to the medical QA study of Ji et al. [2023] and the self-evaluation results of Zhang et al. [2024] places the isolated network firmly in the region $\lambda > q$. Analytically, this is the *invalidation-dominant* regime; numerically, it implies an equilibrium false-statement share of about sixty per cent. Monte-Carlo experiments with our single-network simulator confirm this prediction, ending every run within a few points of the theoretical $\pi_f^{\text{single}} \simeq 0.60$. A model that relies only on its own self-repair routines is therefore destined to retain a majority of invalid statements, in line with the error rates that practitioners see in real deployments.

The picture changes as soon as a second, independent verifier is coupled to the generator. In our formalism that coupling introduces a new hazard d which gives every false statement an additional chance per time-step to be detected and corrected. When d is tuned so that the inequality $\lambda/d < p/(p+q)$ holds, the system crosses a critical boundary and enters a *truth-dominant* regime. Using the parameter values inferred from the two empirical studies and choosing the cross-detection rate d to satisfy $\lambda/d < p/(p+q)$, the dual-network architecture lowers the steady-state false-statement share to about 29 %—a reduction of roughly fifty per cent compared with an isolated network.

Although the mathematics treats d as a single scalar, the concept encompasses several concrete engineering choices. Retrieval-augmented generation raises d by surrounding the model with authoritative passages that expose contradictions. Model-ensemble adjudication raises it further by combining the diverse priors of independently trained models; uncorrelated errors rarely agree, so the aggregate chance that a falsehood slips through diminishes rapidly. Even higher values of d become attainable when a human editor is placed in the loop, although latency and cost then become limiting factors. Our hazard model quantifies these trade-offs: for any target false-statement tolerance one can compute a required detection rate, and thus budget the amount of human or automated scrutiny that must be applied.

The significance of these findings extends beyond artificial systems. Invalidation dynamics of the same mathematical form govern human conversation, peer review and social-media fact-checking. Recognising this shared structure invites a unified research agenda that links network science, cognitive psychology and algorithmic governance. Our future work will explore heterogeneous hazards at the level of individual actors, time-varying detection capacities that respond to workload, and live A/B tests that recover hazard estimates directly from production chat traffic. A deeper ethical analysis will also be required, because the push toward smaller π_f competes with privacy constraints, energy budgets and the carbon footprint of large-scale verification.

In short, the slogan “trust, then verify” admits a precise operational reading: raise the external detection hazard until the inequality $\lambda/d < p/(p+q)$ is satisfied. At that point a network of language models can operate in a truth-dominant regime, even though every individual generator by itself remains prone to invalidation. What began as an abstract Markov model thus delivers a practical design rule for building safer language-technology pipelines.

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